Career Launcher MBA

QA Modern Mathematics

Review Test

The angle of elevation of an aeroplane changes from 60° to 30° in 10 minutes from a point on the ground. Find the speed (in kmph) of the aeroplane if it is flying horizontally at an altitude of 84 km.

(1) 576

(2) 582

(3) 588

(4) 579

Question – 1

The ratio of the sum of odd factors to that of even factors of a natural number is 1: 62. If one of factors of the number is chosen randomly, what is the probability that the chosen factor is an even natural number?

(a) 1/7 (b) 6/7 (c) 1/6 (d) 5/6

Question – 2

If a, b, c are three distinct numbers chosen from numbers 1 to 20, in how many ways can we choose the numbers such that a > b > c and also abc > a + b + c?

(1) 1139

(2) 1140

(3) 570

(4) ${}^{20}C_3 \times 3!$

Question – 3

If there are 10 stations between stations A and B and the train has to stop at station number 3 but will not stop at station number 7, then how many ways are possible to cover the journey from A to B?

(1) 2048

(2) 1024

(3) 512

(4) 256

Question – 4

From a chess board in how many ways can you choose 2 black boxes such that no two of them either belong to same row or same column?

(1) 496

(2) 480

(3) 800

(4) 400

Question – 5

If 5 boys and 5 girls are to be arranged in a row then what is the probability that girls and boys are alternate? Given that no two girls are together.

(1) $\frac{{}^{6}P_{5}}{5!6!}$

(2) $\frac{1}{5}$

(3) $\frac{2}{5}$

(4) $\frac{1}{3}$

Question – 6

There are 100 students in a class who appeared for 3 exams viz. English, Math and Reasoning and 15 of them failed in all 3 subjects. 20 of them passed in all 3 subjects. If 30 of them failed in exactly one subject, then how many of them failed in at least 2 subjects?

(1) 15 (2) 35

(3) 50 (4) 70

Question – 7

In a class, 30 like English, 40 like Maths and 50 like Reasoning. 50 like exactly 2 subjects. How many like all the three subjects if 20 like exactly one subject?

(1) 0

(2) 20

(3) 30

(4) 50

Question – 8

There are 150 students in a class 100 of them like at least 2 subjects of the three, 10 of them like none of the subjects. How many students like at most 1 subject?

(1) 0

(2) 10

(3) 40

(4) 50

Question – 9

In a group of sports lovers, 30 like Cricket, 40 like Football and 50 like Badminton. Find the maximum number of students who may like all the three sports, given that 50 students like at least 2 games.

(1) 0

(2) 20

(3) 30

(4) 50

Question – 10

In how many ways can 20 identical chocolates be distributed among 4 children such that each child receives at least one chocolate and none of them receives 17 chocolates?

(1) 965

(2) 969

(3) ${}^{23}C_3$

(4) ${}^{23}C_3 - 4$

Question – 11

If five different addressed letters are to be put in five different addressed envelopes. In how many ways can you put them if at most 2 of them can go into wrong envelopes?

(1) 11 (2) 10

(3) 60 (4) 59

Question – 12

If 5 identical balls are to be put into 5 boxes such that 2 boxes are of type-A and 3 of them are of type-B. In how many ways can this be done.

(1) 120

(2) 10

(3) ${}^{9}C_{4}$

(4) 25

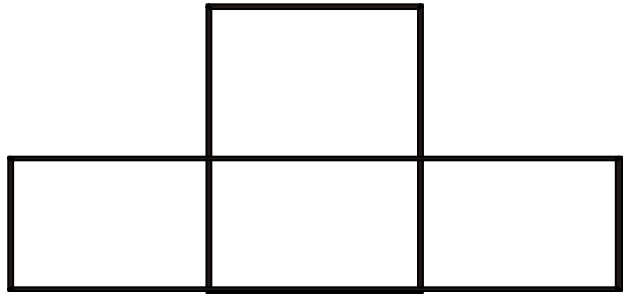
Question – 13

An eight-digit number is formed by using the digits 1 through 8, without repetition. What is the probability that the number thus formed is divisible by 11?

(a) $\frac{3}{35}$ (b) $\frac{2}{35}$ (c) $\frac{4}{35}$ (d) $\frac{6}{35}$

Question – 14

The following tile consists of four identical tiles of unit area each.



If the tiles can be placed in any orientation as long as no two tiles overlap each other, then what is the minimum possible number of such tiles required to obtain a rectangular figure?

(a) 6 (b) 4 (c) 3 (d) 5

Question – 15

What is the minimum value of $+\tan^2\theta+\cot^2\theta+\sec^2\theta+\operatorname{cosec}^2\theta$?

(1) 6

(2) 7

$\sin^2\theta+\cos^2\theta$

(3) 8

(4) 9

Question – 16

If the angles of elevation of a pole from 2 points 120 m apart and on either side of the pole are 30° and 60°, find the height of pole

(1) 30

(2) 40

(3) $30\sqrt{3}$

(4) $40\sqrt{3}$

Question – 17

On a computer screen of size 20 cm × 30 cm, as a screen saver, a circle of radius 7 cm is dancing at random. What is the probability that a point chosen at random is not covered by the circle?

(1) $\frac{1}{2}$

(2) $\frac{1}{4}$

(3) $\frac{1}{7}$

(4) $\frac{7}{100}$

Question – 18

The number of solutions of $x - 10 \sin x = 0$ are

(1) 1

(2) 3

(3) 5

(4) 7

Question – 19

In how many ways can 3 identical colours be painted on 6 faces of a cube such that same colour comes on opposite faces?

(1) 1

(2) 6!

(3) $\frac{6!}{2}$

(4) $3! \times 2$

Question – 20

Thanks!